11 Signal correlation

This chapter covers

- What we mean by signal correlation
- Why you need correlation
- How to implement correlation
- Challenges implementing correlation

Cloud-native systems are *distributed systems*, which refers to many services working together to handle a request. When many services are working together, it turns out it's not always clear which service contributes to a failure or causes latency issues.

The four fundamental observability signal types (logs, metrics, traces, and profiles) each provide insights on their own; however, they typically can't answer all of your questions. Remember, the goal of observability is to derive actionable insights from the signals. So what should you do if a single signal type doesn't yield the insights you hoped for? Use more than one signal type—combine them.

Where might correlation come in handy? Glad you asked:

- *Incident response*—Imagine you're on call for a microservices-based app. You get paged, and the alert contains a link to a Grafana dashboard. From the